

THE TWELFTH POWER MOMENT OF THE RIEMANN-FUNCTION

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1. Introduction

THE classical mean-value theorems for the Riemann Zeta-function, due to Hardy and Littlewood [3] and Ingham [5] respectively, state that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt \sim T \log T, \quad (T \rightarrow \infty), \quad (1)$$

and

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 dt \sim \frac{T}{2\pi^2} (\log T)^4, \quad (T \rightarrow \infty). \quad (2)$$

The problem of establishing similar asymptotic formulae for the mean-value of the sixth or higher powers of $|\zeta(\tfrac{1}{2} + it)|$ appears to be extremely difficult; indeed no such formula has been proved, even on the assumption of the Riemann hypothesis. However for many applications an estimate of the form

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^k dt \ll T^{1+\varepsilon}, \quad (3)$$

where ε is any positive number, is as useful as an asymptotic formula. The estimate (3), for all $k \geq 0$, follows trivially from the Lindelöf hypothesis, (that $\zeta(\tfrac{1}{2} + it) \ll t^\varepsilon$), and so *a fortiori* from the Riemann hypothesis, but no unconditional bound of the form (3) is known, except for $k \leq 4$ when the estimate follows from Ingham's result (2). It is thus natural to ask how good a bound can be given for

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^k dt. \quad (4)$$

It follows from (2) and Kolesnik's estimate [6], namely

$$\zeta(\tfrac{1}{2} + it) \ll t^{\frac{1}{16\sqrt{2}}} (\log t)^{\frac{1}{16\sqrt{2}}}, \quad (t \geq 2), \quad (5)$$

that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^k dt \ll T^{1+173(k-4)/1067} (\log T)^{2k}, \quad (k \geq 4).$$

If we regard this as the 'trivial estimate' for (4), then no non-trivial estimate, for $k > 4$, has been found to date. The object of this paper is to give such an estimate.

THEOREM 1.

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^{12} dt \ll T^2 (\log T)^{17}.$$

As a very simple application of Theorem 1 we note that certain results of Titchmarsh [8], Chapter 12, on the generalized Dirichlet divisor problem, may now be improved. In particular Titchmarsh's Theorem 12.3 (due originally to Hardy and Littlewood), may be sharpened, in the obvious manner, to give

$$\begin{aligned} \Delta_k(x) &\ll x^{k/(4+k)+\varepsilon}, & (4 \leq k \leq 12), \\ \Delta_k(x) &\ll x^{(k-3)/k+\varepsilon}, & (k \geq 12), \end{aligned}$$

the exponent in Titchmarsh's theorem being $(k-1)/(k+2) + \varepsilon$, for $k \geq 4$.

Theorem 1 will be deduced from the following result, which is stronger than Theorem 1 when $|\zeta(\tfrac{1}{2} + it)|$ is sufficiently large.

THEOREM 2. Let $t_1, t_2, \dots, t_R$ be real numbers such that

$$|t_m| \leq T; \quad |t_m - t_n| \geq 1, \quad (m \neq n),$$

and

$$|\zeta(\tfrac{1}{2} + it_m)| \geq V.$$

Then

$$R \ll TV^{-4} (\log T)^5, \quad (V > 0), \quad (6)$$

$$R \ll T^2 V^{-12} (\log T)^{16}, \quad (V > 0), \quad (7)$$

$$R \ll T^5 V^{-32} (\log T)^{40}, \quad (T^{\frac{1}{2}} (\log T)^{\frac{1}{2}} \geq V \geq T^{\frac{1}{2}} (\log T)^{\frac{1}{2}}), \quad (8)$$

$$R \ll TV^{-6} (\log T)^8, \quad (V \geq T^{\frac{1}{2}} (\log T)^{\frac{1}{2}}). \quad (9)$$

We note that (9) yields

$$\zeta(\tfrac{1}{2} + it) \ll t^{\frac{1}{2}} (\log t)^{\frac{1}{2}},$$

so that Theorem 2 almost contains the estimate

$$\zeta(\tfrac{1}{2} + it) \ll t^{\frac{1}{4}} \log t$$

obtained from van der Corput's method, (see Titchmarsh [8], Theorem 12.5).

From Theorem 2 and Kolesnik's estimate (5) we may deduce the following bound for the integral (4).

COROLLARY. *Let $k \geq 15$. Then*

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^k dt \ll T^{1+173(k-6)/1067} (\log T)^{2k}.$$

The asymptotic formulae (1) and (2) are proved by the termwise integration of Dirichlet series corresponding to $|\zeta(\tfrac{1}{2} + it)|^2$ and $|\zeta(\tfrac{1}{2} + it)|^4$ respectively. By careful analysis of this process the formulae (1) and (2) can be replaced by more precise expressions, with error terms $O(T^{346/1067+\epsilon})$ and $O(T^{7/8+\epsilon})$, respectively; details may be found in Balasubramanian [2] and Heath-Brown [4]. However such a method appears to fail completely when we consider powers higher than the fourth. Our treatment, in contrast, is motivated by van der Corput's method; we postpone a description of the ideas involved to the relevant sections.

It will be convenient in the proofs to use the symbol A to denote certain positive numerical constants, not necessarily the same at each occurrence, arising from the $\ll$ (or $O(\dots)$) notation. We shall also use $C_1, C_2, \dots$, to denote positive numerical constants. Other notation will be: $d(n)$ for the number of positive divisors of the natural number n ; $|\mathcal{S}|$ for the number of elements of the set $\mathcal{S}$; $[a, b]$ for the set of real numbers x satisfying $a \leq x \leq b$; γ for Euler's constant; ϵ (in §2 only) may be taken as $\frac{1}{100}$.

2. The preliminary integration

In this section we prove the following mean-value estimate.

LEMMA 1. *Let $(\log T)^2 \leq G \leq T^{\frac{1}{4}}$. Then*

$$\begin{aligned} \int_{T-G}^{T+G} |\zeta(\tfrac{1}{2} + it)|^2 dt &\ll G \log T + G \sum_{T^{1/3} \leq K \leq 2^k \leq TG^{-2}(\log T)^2} (TK)^{-1} (|S(K)| \\ &\quad + K^{-1} \int_0^K |S(x)| dx) e^{-G^3 K/T}, \end{aligned}$$

where

$$S(x) = S(x, K, T) = \sum_{K < n \leq K+x} (-1)^n d(n) \exp(ief(T, n))$$

and

$$f(T, n) = 2T \sinh^{-1}((\pi n/2T)^{1/2}) + (\pi^2 n^2 + 2\pi nT)^{1/2}.$$

Lemma 1 already contains the most important new ideas in the proof of our results. The estimate should be regarded as a precise form of van der Corput's method applied to $\zeta(s)$. The usual form of van der Corput's inequality states that

$$\left| \sum_{1 \leq n \leq N} \exp(ief(n)) \right|^2 \leq 2N^2 H^{-1} + 4NH^{-1} \sum_{0 < h < H} (1 - h/H) \sum_{1 \leq n \leq N-h} \cos(f(n+h) - f(n)). \quad (10)$$

However one may, in general, look on the inequality as bounding a mean-value; for by Lemma 1.9 of Montgomery [7], (a result due to Gallagher), we have

$$\int_{1-\delta}^{1+\delta} \left| \sum_{1 \leq n \leq N} \exp(ief(n)) \right|^2 dt \ll \delta \sum \exp(i(f(m) - f(n)))(1 - \delta |f(m) - f(n)|), \quad (11)$$

where the sum on the right is for $1 \leq m, n \leq N$, $\delta |f(m) - f(n)| \leq 1$; and this corresponds to the inequality (10) with δ for N/H and $N |f(n+h) - f(n)|$ for h . Then Lemma 1 corresponds to (11) with the double exponential sum 'dualized' by taking the Fourier transform with respect to n , for fixed $h = m - n$, to produce $S(x)$, the latter process being the one given by Theorem 4.9 of Titchmarsh [8]. The procedure sketched above would be difficult to carry out in practice, and we save a considerable amount of work by appealing to the following theorem of Atkinson [1], which already contains the important features of Lemma 1.

Let $T \geq 2$, $T/10 \leq N \leq 10T$,

$$N' = N'(T) = \frac{T}{2\pi} + \frac{N}{2} - \left(\frac{NT}{4\pi} + \frac{N^2}{4} \right)^{\frac{1}{2}}.$$

Then

$$\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \left(\log \frac{T}{2\pi} + 2\gamma - 1 \right) + \Sigma_1(T) + \bar{\Sigma}_1(T) + \Sigma_2(T) + E(T), \quad (12)$$

where

$$\begin{aligned}\Sigma_1(T) &= 2^{-1/2} \sum_{n \leq N} (-1)^n d(n) n^{-1/2} \left(\frac{T}{2\pi n} + \frac{1}{4} \right)^{-1/2} \\ &\quad \times \left(\sinh^{-1} \left(\left(\frac{\pi n}{2T} \right)^{1/2} \right) \right)^{-1} \sin \left(\frac{\pi}{4} + f(T, n) \right), \\ \Sigma_2(T) &= -2 \sum_{n \leq N'} d(n) n^{-1/2} \left(\log \frac{T}{2\pi n} \right)^{-1} \sin(g(T, n)), \\ g(T, n) &= T \log \frac{T}{2\pi n} - T - \frac{\pi}{4}, \\ E(T) &\ll (\log T)^2.\end{aligned}$$

This theorem will be the starting point for the proof of Lemma 1. Let $H = G \log T$ and

$$I(t) = \int_0^t |\zeta(\tfrac{1}{2} + iu)|^2 du;$$

then integration by parts yields

$$\int_{-H}^H |\zeta(\tfrac{1}{2} + i(T+t))|^2 e^{-t^2/G^2} dt = 2 \int_{-H}^H I(T+t) e^{-t^2/G^2} t G^{-2} dt + O(1), \quad (13)$$

since $I(T \pm H) \ll T^2$, by (1). We now apply the formula (12) with $N = T$. Then

$$2 \int_{-H}^H E(T+t) e^{-t^2/G^2} t G^{-2} dt \ll (\log T)^2. \quad (14)$$

Σ_2 also makes a negligible contribution, as we proceed to show. We have $N'(T+t) = N'(T) + O(|t|)$, so that

$$\Sigma_2(T+t) = -2 \sum_{n \leq N'(T)} d(n) n^{-1/2} \left(\log \frac{(T+t)}{2\pi n} \right)^{-1} \sin(g(T+t, n)) + O(HT^{-1/2});$$

here we use the observation that

$$T \ll N'(T) \leq \left(\frac{3}{4} \right) \left(\frac{T}{2\pi} \right),$$

whence

$$\left(\log \frac{(T+t)}{2\pi n}\right)^{-1} \ll 1.$$

Moreover

$$\left(\log \frac{(T+t)}{2\pi n}\right)^{-1} = \left(\log \frac{T}{2\pi n}\right)^{-1} + O\left(\frac{|t|}{T}\right),$$

and, writing $g = g(T, n)$, $g' = g'(T, n)$, $g'' = g''(T, n)$, we have by Taylor's Theorem both

$$g(T+t, n) = g + tg' + \frac{t^2 g''}{2} + O(|t|^3 T^{-2})$$

and

$$\sin(g(T+t, n)) = \sin\left(g + tg' + \frac{t^2 g''}{2}\right) + O(|t|^3 T^{-2}).$$

Hence

$$\Sigma_2(T+t) = -2 \sum_{n \leq N(T)} d(n) n^{-\frac{1}{2}} \left(\log \frac{T}{2\pi n}\right)^{-1} \sin\left(g + tg' + \frac{t^2 g''}{2}\right) + O(HT^{-\frac{1}{2}}), \quad (15)$$

since $|t| \leq H \ll T^{\frac{1}{2}}$. Thus to calculate the contribution of Σ_2 to (13) we consider

$$\int_{-H}^H e^{itg' + it^2 g''/2 - t^2/G^2} t G^{-2} dt = \int_{-\infty}^{\infty} e^{itg' + it^2 g''/2 - t^2/G^2} t G^{-2} dt + O(e^{-H^2/G^2}). \quad (16)$$

To evaluate the integral on the right we use the formula

$$\int_{-\infty}^{\infty} t e^{\alpha t - \beta t^2} dt = \frac{1}{2} \pi^{\frac{1}{2}} \alpha \beta^{-\frac{3}{2}} \exp\left(\frac{\alpha^2}{4\beta}\right), \quad (\operatorname{Re}(\beta) > 0),$$

whence

$$\int_{-\infty}^{\infty} e^{itg' + it^2 g''/2 - t^2/G^2} t G^{-2} dt \ll \exp(-AG^2), \quad (17)$$

since $g' \gg 1$, $g'' \ll T^{-1}$ and $(\log T)^2 \ll G \ll T^{\frac{1}{2}}$. The estimates (15), (16) and

(17) now yield

$$2 \int_{-H}^H \Sigma_2(T+t) e^{-t^2/G^2} t G^{-2} dt \ll HT^{s-1} + \sum_{n \leq AT} d(n) n^{-1} \exp(-(\log T)^2) \ll 1. \quad (18)$$

There remains the term

$$2 \int_{-H}^H \left((T+t) \left(\log \frac{(T+t)}{2\pi} + 2\gamma - 1 \right) + \Sigma_1(T+t) \right) e^{-t^2/G^2} t G^{-2} dt.$$

This we integrate by parts to give

$$\int_{-H}^H \left(\log \frac{(T+t)}{2\pi} + 2\gamma \right) e^{-t^2/G^2} dt + \int_{-H}^H \Sigma_1'(T+t) e^{-t^2/G^2} dt + O(1); \quad (19)$$

here we have used the trivial estimate

$$(T \pm H) \left(\log \frac{(T \pm H)}{2\pi} + 2\gamma - 1 \right) + \Sigma_1(T \pm H) \ll T^2 \ll e^{H^2/G^2}$$

to obtain the error term $O(1)$. For the first integral in (19) we have immediately

$$\int_{-H}^H \left(\log \frac{(T+t)}{2\pi} + 2\gamma \right) e^{-t^2/G^2} dt \ll G \log T. \quad (20)$$

To bound the second integral in (19) we must first replace $\Sigma_1'(T+t)$ by a simpler expression. We note that $f'(t, n) = 2 \sinh^{-1}((\pi n/2t)^{1/2})$, whence

$$\begin{aligned} \frac{d}{dt} \left(\left(\frac{t}{2\pi n} + \frac{1}{4} \right)^{-1/2} \left(\sinh^{-1} \left(\left(\frac{\pi n}{2t} \right)^{1/2} \right) \right)^{-1} \sin \left(\frac{\pi}{4} + f(t, n) \right) \right) \\ = 2 \left(\frac{t}{2\pi n} + \frac{1}{4} \right)^{-1/2} \cos \left(\frac{\pi}{4} + f(t, n) \right) + O(T^{-1/2} n^{-1/2}), \end{aligned}$$

for $|t - T| \leq H$. Moreover, writing $f = f(T, n)$, $f' = f'(T, n)$, $f'' = f''(T, n)$, we have, by Taylor's Theorem,

$$\begin{aligned} \left(\frac{(T+t)}{2\pi n} + \frac{1}{4} \right)^{-1/2} \cos \left(\frac{\pi}{4} + f(T+t, n) \right) &= \left(\frac{T}{2\pi n} + \frac{1}{4} \right)^{-1/2} \cos \left(\frac{\pi}{4} + f + tf + \frac{t^2 f''}{2} \right) \\ &+ O(|t| n^{1/2} T^{-1/2}) + O(|t|^3 n^{1/2} T^{-3/2}). \end{aligned}$$

Hence

$$\begin{aligned} \Sigma'_1(T+t) = 2^{\frac{1}{2}} \sum_{n \leq T} (-1)^n d(n) n^{-\frac{1}{2}} \left(\frac{T}{2\pi n} + \frac{1}{4} \right)^{-\frac{1}{2}} \cos \left(\frac{\pi}{4} + f + tf + \frac{t^2 f''}{2} \right) \\ + O \left(\sum_{n \leq T} d(n) n^{-\frac{1}{2}} (T^{-\frac{1}{2}} n^{-\frac{1}{2}} + |t| n^{\frac{1}{2}} T^{-\frac{1}{2}} + |t|^3 n^{\frac{1}{2}} T^{-\frac{1}{2}}) \right). \end{aligned}$$

We readily see that the error term above is

$$\ll T^{-1/2+\varepsilon} + |t| T^{-1/2+\varepsilon} + |t|^3 T^{-3/2+\varepsilon} \ll 1$$

for $|t| \leq H$; the contribution to (19) is therefore

$$\ll G \log T. \quad (21)$$

We must now consider

$$\int_{-H}^H e^{itf' + it^2 f''/2 - t^2/G^2} dt = \int_{-\infty}^{\infty} e^{itf' + it^2 f''/2 - t^2/G^2} dt + O(e^{-H^2/G^2}). \quad (22)$$

To evaluate the integral on the right we use the formula

$$\int_{-\infty}^{\infty} e^{\alpha t - \beta t^2} dt = \left(\frac{\pi}{\beta} \right)^{\frac{1}{2}} \exp \left(\frac{\alpha^2}{4\beta} \right), \quad (\operatorname{Re}(\beta) > 0),$$

which shows that

$$\int_{-\infty}^{\infty} e^{itf' + it^2 f''/2 - t^2/G^2} dt \ll \exp(-(\log T)^2) \quad (23)$$

for $n \geq TG^{-2}(\log T)^3$, and that

$$\int_{-\infty}^{\infty} e^{itf' + it^2 f''/2 - t^2/G^2} dt = \pi^{\frac{1}{2}} G \exp \left(\frac{-G^2 f'^2}{4} \right) + O(G^3 n^{\frac{1}{2}} T^{-\frac{1}{2}+\varepsilon}) \quad (24)$$

for $n \leq 2TG^{-2}(\log T)^3$. The error term in (22) and the integral (23) contribute a total

$$\ll \exp(-(\log T)^2) \sum_{n \leq T} d(n) n^{-\frac{1}{2}} T^{-\frac{1}{2}} \ll 1 \quad (25)$$

to (19), and the error term of (24) contributes

$$\ll \sum_{n \leq T} d(n) G^3 n^{\frac{1}{2}} T^{-\frac{1}{2}+\varepsilon} \ll G^3 T^{-1+2\varepsilon} \ll G \quad (26)$$

to (19). The estimates (13), (14), (18), (20), (21), (24), (25) and (26) yield

$$\int_{-H}^H |\zeta(\tfrac{1}{2} + i(T+t))|^2 e^{-t^2/G^2} dt = (2\pi)^{\frac{1}{2}} G \sum_n (-1)^n d(n) n^{-\frac{1}{2}} \left(\frac{T}{2\pi n} + \frac{1}{4} \right)^{-\frac{1}{2}} \\ \times \cos \left(\frac{\pi}{4} + f(T, n) \right) \exp \left(-G^2 \left(\sinh^{-1} \left(\left(\frac{\pi n}{2T} \right)^{\frac{1}{2}} \right) \right)^2 \right) + O(G \log T).$$

where the summation is for $n \leq 2^k$, $2^{k+1} \leq TG^{-2} (\log T)^3 < 2^k$. We now observe that

$$\sum_{K < n \leq 2K} (-1)^n d(n) n^{-\frac{1}{2}} \left(\frac{T}{2\pi n} + \frac{1}{4} \right)^{-\frac{1}{2}} \\ \times \exp \left(if(T, n) - G^2 \left(\sinh^{-1} \left(\left(\frac{\pi n}{2T} \right)^{\frac{1}{2}} \right) \right)^2 \right) \\ \ll (TK)^{-\frac{1}{2}} \left(|S(K)| + K^{-1} \int_0^K |S(x)| dx \right) e^{-G^2 K/T},$$

by partial summation. Moreover $S(x) \ll K \log T$ trivially, whence

$$G \sum_{K-2^k \leq T^{1/2}} (TK)^{-\frac{1}{2}} \left(|S(K)| + K^{-1} \int_0^K |S(x)| dx \right) e^{-G^2 K/T} \ll G \log T.$$

Lemma 1 now follows, since

$$\int_{-G}^G |\zeta(\tfrac{1}{2} + i(T+t))|^2 dt \ll \int_{-H}^H |\zeta(\tfrac{1}{2} + i(T+t))|^2 e^{-t^2/G^2} dt.$$

3. Large values of $S(x, K, T)$

In this section we apply Halász' method to investigate the frequency of large values of $S(x, K, T)$.

LEMMA 2. Let $\mathcal{S}$ be a set of real numbers t , such that

$$\frac{T}{2} \leq t \leq T; \quad |t - t_r| \leq J; \quad |t - t_s| \geq G, \quad (r \neq s).$$

Then, if $K \leq T/(\log T)$,

$$\sum_{t \in \mathcal{S}} |S(x, K, t)|^2 \ll K(\log T)^3 (K + G^{-1}(KT)^{\frac{1}{2}} \log T + \lambda |\mathcal{S}|),$$

where

$$\lambda = J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}}, \quad \left(0 < J \leq \text{Min} \left(K T^{\frac{1}{2}}, \frac{T}{2} \right)\right),$$

$$\lambda = J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}}, \quad \left(K T^{\frac{1}{2}} < J \leq \frac{T}{2} \right).$$

Our starting point is the inequality

$$\sum_r |(\xi, \phi_r)|^2 \leq \|\xi\|^2 \text{Max}_r \sum_s |(\phi_r, \phi_s)|, \quad (27)$$

valid in any inner product space; this is due to Bombieri, and may be found in Montgomery [7], Lemma 1.5. We apply this to the space of finite complex-valued sequences, with the inner product

$$((a_n), (b_n)) = \sum a_n \bar{b}_n.$$

We take ξ to be the sequence whose n -th element is $(-1)^n d(n)$ for $K < n \leq K+x$, and 0 otherwise, and ϕ_r to be the sequence whose n -th element is $\exp(i f(t_r, n))$ for $K < n \leq 2K$, and 0 otherwise. It follows from (27) that

$$\sum_{t \in \mathcal{S}} |S(x, K, t)|^2 \leq \left(\sum_{K < n \leq K+x} (d(n))^2 \right) \times \text{Max}_r \sum_s \left| \sum_{K < n \leq 2K} \exp(i(f(t_r, n) - f(t_s, n))) \right|. \quad (28)$$

We must now bound

$$S = \sum_{K < n \leq 2K} \exp(i f(n)),$$

where $f(n) = f(t_r, n) - f(t_s, n)$. We shall use only the simplest techniques; it is therefore quite possible that Lemma 2, and hence Theorem 2, might be sharpened by giving more refined estimates for S , but the improvement would probably be very small.

We shall require estimates for the derivatives $f^{(j)}(u)$. We write

$$g(z) = \sinh^{-1}(z) + z(z^2 + 1)^{\frac{1}{2}}.$$

Then $g(z)$, regarded as a function of a complex variable, is regular for $|z| < 1$, and may therefore be expanded as a power series

$$g(z) = \sum_{m=0}^{\infty} a_m z^m, \quad (a_0 = 0, a_1 = 2),$$

with

$$a_m \ll 2^m. \quad (29)$$

Thus

$$f(u) = 2t_r g\left(\left(\frac{\pi u}{2t_r}\right)^{\frac{1}{2}}\right) - 2t_s g\left(\left(\frac{\pi u}{2t_s}\right)^{\frac{1}{2}}\right) = \sum_{m=1}^{\infty} 2a_m \left(t_r \left(\frac{\pi}{2t_r}\right)^{m/2} - t_s \left(\frac{\pi}{2t_s}\right)^{m/2}\right) u^{m/2}.$$

By (29) we may differentiate termwise l times, if $u \leq 2K$ and

$$4\left(\frac{\pi}{2t_r}\right)[2K], 4\left(\frac{\pi}{2t_s}\right)(2K) < 1;$$

and this conditions holds, for large enough T , since $K \leq T/(\log T)$. From the First Mean-Value Theorem we have

$$t_r^{1-m/2} - t_s^{1-m/2} = \left(1 - \frac{m}{2}\right)(t_r - t_s)t^{-m/2}$$

for some t between t_r and t_s , whence

$$\left|1 - \frac{m}{2}\right| \Delta T^{-m/2} \leq |t_r^{1-m/2} - t_s^{1-m/2}| \leq m \Delta \left(\frac{T}{2}\right)^{-m/2}, \quad (30)$$

where $\Delta = |t_r - t_s|$. We now have, by (29) and (30), if $K \leq u \leq 2K$,

$$f^{(l)}(u) = a_1(2\pi)^{\frac{1}{2}} C_l (t_r^{\frac{1}{2}} - t_s^{\frac{1}{2}}) u^{1/2-l} + O\left(\Delta K^{-l} \sum_{m=2}^{\infty} 4^m \left(\frac{K}{T}\right)^{m/2}\right), \quad (31)$$

for some numerical constant C_l . By (30) with $m = 1$ we have

$$\Delta K^{1/2-l} T^{-\frac{1}{2}} \ll |a_1(2\pi)^{\frac{1}{2}} C_l (t_r^{\frac{1}{2}} - t_s^{\frac{1}{2}}) u^{\frac{1}{2}-l}| \ll \Delta K^{1/2-l} T^{-\frac{1}{2}},$$

for $K \leq u \leq 2K$, whereas the error term of (31) is $\ll \Delta K^{1-l} T^{-1}$. Since $K \ll T/(\log T)$ we conclude that

$$\Delta K^{1/2-l} T^{-\frac{1}{2}} \ll |f^{(l)}(u)| \ll \Delta K^{1/2-l} T^{-\frac{1}{2}}. \quad (32)$$

We are now ready to estimate S . We first apply Lemma 4.8 of Titchmarsh [8], which yields

$$S = \int_K^{2K} \exp(if(u)) du + O(1),$$

providing that

$$\Delta \leq A(KT)^{\frac{1}{2}}. \quad (33)$$

Lemma 4.2 of [8] then yields

$$S \ll \Delta^{-1}(KT)^{\frac{1}{2}}. \quad (34)$$

Secondly we may apply Theorem 5.9 of [8], whence, by (32) with $l = 2$,

$$S \ll \Delta^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} + \Delta^{-\frac{1}{2}} K^{\frac{3}{2}} T^{\frac{1}{2}}. \quad (35)$$

Finally we may use Theorem 5.11 of [8], which yields, by (32) with $l = 3$,

$$S \ll \Delta^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} + \Delta^{-\frac{1}{2}} K^{\frac{3}{2}} T^{\frac{1}{2}}. \quad (36)$$

The right hand side of (28) can now be estimated. For those terms for which (33) holds we have

$$\sum_s \left| \sum_{K < n \leq 2K} \exp(i(f(t_s, n) - f(t_s, n))) \right| \ll (KT)^{\frac{1}{2}} \sum_s |t_s - t_s|^{-1},$$

where the summations are for s satisfying $0 < |t_s - t_s| \leq A(KT)^{\frac{1}{2}}$. From the condition $|t_s - t_{s'}| \geq G$, ($s \neq s'$), we have

$$\sum_s |t_s - t_s|^{-1} \ll G^{-1} \log T,$$

whence

$$\sum_{s: \Delta \leq A(KT)^{1/2}} \left| \sum_{K < n \leq 2K} \exp(i(f(t_s, n) - f(t_s, n))) \right| \ll K + G^{-1}(KT)^{\frac{1}{2}} \log T, \quad (37)$$

since the term $s = r$ contributes $\ll K$. In case $A(KT)^{\frac{1}{2}} \leq \Delta \leq KT^{\frac{1}{2}}$ we use (35), which immediately yields

$$\sum_s \left| \sum_{K < n \leq 2K} \exp(i(f(t_s, n) - f(t_s, n))) \right| \ll (\text{Min}(J, KT^{\frac{1}{2}}))^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} |\mathcal{S}|, \quad (38)$$

and for $\Delta \geq KT^{\frac{1}{2}}$ we use (36), whence

$$\sum_s \left| \sum_{K < n \leq 2K} \exp(i(f(t_s, n) - f(t_s, n))) \right| \ll J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} |\mathcal{S}|. \quad (39)$$

The estimates (37), (38) and (39) yield

$$\sum_s \left| \sum_{K < n \leq 2K} \exp(i(f(t_s, n) - f(t_s, n))) \right| \ll K + G^{-1}(KT)^{\frac{1}{2}} \log T + \lambda |\mathcal{S}|,$$

where the summation is now over all values of s , and Lemma 2 follows from (28), on noting that

$$\sum_{K < n \leq K+x} (d(n))^2 \ll K (\log T)^3.$$

4. Preparation for the proof of Theorem 2

In order to relate the discrete set occurring in Theorem 2 to the integral in Lemma 1, we shall require the following lemma.

LEMMA 3. Let $t \geq 10$. Then

$$|\zeta(\tfrac{1}{2} + it)|^2 \ll (\log t) \left(1 + \int_{-(\log t)^2}^{(\log t)^2} e^{-|u|} |\zeta(\tfrac{1}{2} + i(t+u))|^2 du \right).$$

Let $s = \tfrac{1}{2} + \delta + i\tau$, where $\delta = (\log t)^{-1}$, $t/2 \leq \tau \leq 3t/2$; then, by termwise integration we have

$$\frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} (\zeta(s+w))^2 \Gamma(w) dw = \sum_1^{\infty} d(n) n^{-s} e^{-n} \ll 1.$$

The integrand has poles at $w=0$ and $w=1-s$; at $w=0$ the residue is $(\zeta(s))^2$, and at $w=1-s$ the residue is $O(1)$. Thus, moving the line of integration to $\operatorname{Re}(w) = -\delta$, we have

$$|\zeta(s)|^2 \ll 1 + \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i(\tau+v))|^2 e^{-|v|} \frac{dv}{(\delta + |v|)}, \quad (40)$$

since

$$\Gamma(\pm\delta + iv) \ll e^{-|v|} (\delta + |v|)^{-1}. \quad (41)$$

Moreover, by the functional equation for $\zeta(s)$,

$$|\zeta(\tfrac{1}{2} - \delta + i\tau)|^2 \ll \tau^{2\delta} |\zeta(\tfrac{1}{2} + \delta + i\tau)|^2,$$

so that (40) holds for $s = \tfrac{1}{2} - \delta + i\tau$ also. By the residue theorem we have

$$(\zeta(s_0))^2 = \frac{1}{2\pi i} \int_P (\zeta(s_0+z))^2 \Gamma(z) dz,$$

where $s_0 = \tfrac{1}{2} + it$, and the path P is the rectangle with corners $s_0 \pm \delta \pm i(\log t)^2$, described in the positive direction. Since

$$(\zeta(s_0+z))^2 \ll t, \quad \Gamma(z) \ll \exp(-(\log t)^2),$$

on the two shorter sides, the corresponding contribution is $O(1)$; and the two longer sides contribute

$$\ll \int_{-(\log t)^2}^{(\log t)^2} e^{-|w|} \left(1 + \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i(t+w+v))|^2 e^{-|v|} \frac{dv}{(\delta + |v|)} \right) \frac{dw}{(\delta + |w|)}, \quad (42)$$

by (40) and (41). We have trivially

$$\int_{-(\log t)^2}^{(\log t)^2} e^{-|w|} \frac{dw}{(\delta + |w|)} \ll \delta^{-1} = \log t,$$

and for the remaining part of (42) we have, after changing the order of integration,

$$\begin{aligned} &\ll \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i(t+u))|^2 \left(\int_{-\infty}^{\infty} e^{-|w| - |u-w|} (\delta + |w|)^{-1} (\delta + |u-w|)^{-1} dw \right) du \\ &\ll \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i(t+u))|^2 e^{-|u|} \delta^{-1} du. \end{aligned}$$

To complete the proof of Lemma 3 we have only to note that

$$\left(\int_{-\infty}^{-(\log t)^2} + \int_{(\log t)^2}^{\infty} \right) |\zeta(\tfrac{1}{2} + i(t+u))|^2 e^{-|u|} \delta^{-1} du \ll 1,$$

since $|\zeta(\tfrac{1}{2} + i(t+u))|^2 \ll 1 + |t+u|$.

We now let $\mathcal{S}_1$ be a set of points satisfying the conditions of Theorem 2, but lying in the interval $[T_0, T_0 + J]$, where $2T/3 \leq T_0$ and $T_0 + J \leq 5T/6$. We define G by

$$BG(\log T)^2 = V^2, \quad (43)$$

where B is a positive numerical constant which we shall specify later. Let

$$\mathcal{S}'_1 = \mathcal{S}'_1(\tau) = \mathcal{S}_1 \cap \left[\tau - \frac{G}{2}, \tau + \frac{G}{2} \right].$$

Then by Lemma 3

$$V^2 |\mathcal{S}'_1| \leq C_1 (\log T) \left(|\mathcal{S}'_1| + \int_{\tau-G}^{\tau+G} |\zeta(\tfrac{1}{2} + it)|^2 \left(\sum_{t_n \in \mathcal{S}'_1} e^{-|t-t_n|} \right) dt \right), \quad (44)$$

providing that $G/2 + (\log(\tau + G))^2 \leq G$; a sufficient condition is

$$G \geq (\log T)^3. \quad (45)$$

Now since $|t_m - t_n| \geq 1$ for $m \neq n$,

$$\sum e^{-|t-t_n|} \ll 1,$$

and (44) simplifies to

$$BG (\log T)^2 |\mathcal{S}'_1| \leq C_2 (\log T) \left(|\mathcal{S}'_1| + \int_{\tau-G}^{\tau+G} |\zeta(\tfrac{1}{2} + it)|^2 dt \right).$$

If we assume that $7T/12 \leq \tau \leq 11T/12$, then Lemma 1 applies, providing that

$$(\log T)^3 \leq G \leq \tfrac{1}{2} T^{\frac{1}{2}}; \quad (46)$$

this includes the condition (45). Thus, writing

$$\begin{aligned} \sum(\tau) = G \sum_{\tfrac{1}{2}T^{1/3} \leq K = 2^k \leq TG^{-2}(\log T)^3} (TK)^{-1} \\ \times \left(|S(K, K, \tau)| + K^{-1} \int_0^K |S(x, K, \tau)| dx \right) e^{-G^2 K/T}, \end{aligned}$$

we have

$$BG (\log T)^2 |\mathcal{S}'_1| \leq C_3 (\log T) (|\mathcal{S}'_1| + G \log T + \sum(\tau)).$$

We choose $B = 2C_3$, whence

$$G (\log T) |\mathcal{S}'_1| \leq |\mathcal{S}'_1| + \sum(\tau).$$

Thus, if $(\log T)^4 \geq 2$,

$$|\mathcal{S}'_1(\tau)| \leq 2(G \log T)^{-1} \sum(\tau).$$

Cauchy's inequality yields

$$\begin{aligned} \left(\sum(\tau) \right)^2 \leq G^2 T^{-1} \left(\sum_K K^{\frac{1}{2}} e^{-G^2 K/T} \right) \\ \times \left\{ \sum_K K^{-\frac{1}{2}} \left(|S(K, K, \tau)|^2 + K^{-1} \int_0^K |S(x, K, \tau)|^2 dx \right) e^{-G^2 K/T} \right\}, \end{aligned}$$

and, since

$$\sum_K K^\alpha e^{-G^2 K/T} \ll \left(\frac{T}{G^2} \right)^\alpha, \quad (47)$$

for any constant $\alpha > 0$, we have

$$|\mathcal{S}'_1(\tau)| \leq |\mathcal{S}'_1(\tau)|^2 \leq 4(G \log T)^{-2} \left(\sum(\tau) \right)^2 \ll G^{-1} T^{-1} (\log T)^{-2} \sum_K K^{-1} \\ \times \left(|S(K, K, \tau)|^2 + K^{-1} \int_0^K |S(x, K, \tau)|^2 dx \right) e^{-G^2 K/T}. \quad (48)$$

We now let $\mathcal{S}_2$ run over numbers $\tau = T_0 + G/2 + nG$, where $\mathcal{S}'_1(\tau) \neq \emptyset$, and the integer n is such that $T_0 \leq \tau \leq T_0 + J + G/2$. Lemma 2 applied to $\mathcal{S}_2$, and the estimate (48), then yield

$$\sum_{\tau \in \mathcal{S}_2} |\mathcal{S}'_1(\tau)| \ll G^{-1} T^{-1} (\log T) \sum_K K^{\frac{1}{2}} (K + G^{-1} (KT)^{\frac{1}{2}} \log T + \lambda_K |\mathcal{S}_2|) e^{-G^2 K/T}, \quad (49)$$

where λ_K is the value of λ corresponding to K . By (47) we have immediately

$$G^{-1} T^{-1} (\log T) \sum_K K^{\frac{1}{2}} (K + G^{-1} (KT)^{\frac{1}{2}} \log T) e^{-G^2 K/T} \ll TG^{-3} (\log T)^2. \quad (50)$$

To estimate

$$\sum = \sum_K K^{\frac{1}{2}} \lambda_K e^{-G^2 K/T}$$

we consider two cases. According to Lemma 2 we have $\lambda_K \leq J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}}$ for all K . Hence, by (47),

$$\sum \leq \sum_K J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} e^{-G^2 K/T} \ll J^{\frac{1}{2}} T^{\frac{1}{2}} G^{-1}. \quad (51)$$

Now suppose that $JT^{-1} \geq T/G^2$. Then, by Lemma 2,

$$\lambda_K \leq J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}}$$

for $K \leq JT^{-1}$, and by (47),

$$\sum_{K \leq JT^{-1/2}} J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}} e^{-G^2 K/T} \ll J^{\frac{1}{2}} T^{\frac{1}{2}} G^{-1}. \quad (52)$$

For $JT^{-1} \leq K \leq TG^{-2} (\log T)^3$ Lemma 2 gives $\lambda_K \leq J^{\frac{1}{2}} K^{\frac{1}{2}} T^{-\frac{1}{2}}$, whence

$$\sum_{K \geq J/T^{1/2}} K^{\frac{1}{2}} \lambda_K e^{-G^2 K/T} \ll JT^{-1} \exp(-G^2 J/T^{\frac{3}{2}}) \ll J^{\frac{1}{2}} T^{\frac{1}{2}} G^{-1}, \quad (53)$$

since

$$\sum_{K \geq K_0} K^{\frac{1}{2}} e^{-G^2 K/T} \ll K^{\frac{1}{2}} e^{-G^2 K_0/T}$$

for $K_0 \geq T/G^2$, and $e^{-\beta} \ll \beta^{-\frac{1}{2}}$ for $\beta = G^2 J/T^{\frac{1}{2}} \geq 1$. From (52) and (53) we deduce

$$\sum \ll J^{\frac{1}{2}} T^{\frac{1}{2}} G^{-\frac{1}{2}}$$

when $J \geq T^{\frac{1}{2}} G^{-2}$. In conjunction with (49), (50) and (51) this yields

$$\sum_{\tau \in \mathcal{S}_2} |\mathcal{S}'_1(\tau)| \leq C_4 (TG^{-3} (\log T)^2 + \mu |\mathcal{S}_2|),$$

where

$$\begin{aligned} \mu &= J^{\frac{1}{2}} G^{-\frac{1}{2}} \log T, & \left(0 < J \leq \min \left(T^{\frac{1}{2}} G^{-2}, \frac{T}{2} \right) \right), \\ \mu &= J^{\frac{1}{2}} G^{-\frac{1}{2}} T^{\frac{1}{2}} \log T, & \left(T^{\frac{1}{2}} G^{-2} \leq J \leq \frac{T}{2} \right). \end{aligned}$$

By definition of $\mathcal{S}_2$ we have

$$|\mathcal{S}_2| \leq \sum_{\tau \in \mathcal{S}_2} |\mathcal{S}'_1(\tau)|.$$

Hence if

$$\mu C_4 \leq \frac{1}{2} \tag{54}$$

then

$$|\mathcal{S}_1| \leq \sum_{\tau \in \mathcal{S}_2} |\mathcal{S}'_1(\tau)| \ll TG^{-3} (\log T)^2. \tag{55}$$

The condition (54) will be satisfied for

$$J \leq C_5 G^3 (\log T)^{-2}, \tag{56}$$

or for

$$T^{\frac{1}{2}} G^{-2} \leq J \leq C_6 G^{13} T^{-3} (\log T)^{-6}. \tag{57}$$

5. Proofs of Theorems 1, 2 and the corollary

We first prove Theorem 2. Take $\mathcal{S}_3$ to be any set satisfying the conditions of Theorem 2, but contained in $[2T/3, 5T/6]$. We divide the interval into N subintervals of length $J = T/6N$, and $\mathcal{S}_3$ into N subsets satisfying the

conditions for $\mathcal{S}_1$, one contained in each subinterval. Then, applying (55) to each subset, we have

$$|\mathcal{S}_3| \ll NTG^{-3}(\log T)^2, \quad (58)$$

providing that

$$\frac{T}{6N} \leq C_5 G^3 (\log T)^{-2}, \quad (59)$$

or

$$T^{\frac{1}{3}} G^{-2} \leq \frac{T}{6N} \leq C_6 G^{13} T^{-3} (\log T)^{-6}. \quad (60)$$

We consider two cases. Suppose firstly that

$$T^{\frac{1}{3}} G^{-2} \geq \frac{1}{2} C_6 G^{13} T^{-3} (\log T)^{-6}.$$

In this case we define N by

$$\frac{T}{6N} \leq C_5 G^3 (\log T)^{-2} < \frac{T}{6(N-1)},$$

so that (59) holds. Then

$$N \ll 1 + TG^{-3} (\log T)^2$$

and (58) yields

$$|\mathcal{S}_3| \ll TG^{-3} (\log T)^2 + T^2 G^{-6} (\log T)^4. \quad (61)$$

Alternatively let $T^{\frac{1}{3}} G^{-2} \leq \frac{1}{2} C_6 G^{13} T^{-3} (\log T)^{-6}$, that is

$$G \geq C_7 T^{\frac{1}{6}} (\log T)^{\frac{1}{3}}. \quad (62)$$

We now define N by

$$\frac{T}{6N} \leq C_6 G^{13} T^{-3} (\log T)^{-6} < \frac{T}{6(N-1)}.$$

Then, if $N \geq 2$, we have

$$\frac{T}{6N} \geq \frac{T}{12(N-1)} > \frac{1}{2} C_6 G^{13} T^{-3} (\log T)^{-6} > T^{\frac{1}{3}} G^{-2},$$

while if $N = 1$, $T/6N > T^{\frac{1}{3}} G^{-2}$ by (62). Thus (60) holds, and (58) may be applied with

$$N \ll 1 + T^4 G^{-13} (\log T)^6;$$

this yields

$$|\mathcal{S}_3| \ll TG^{-3} (\log T)^2 + T^5 G^{-16} (\log T)^8. \quad (63)$$

Using (43) we see that (61) and (63) imply the estimates (7), (8) and (9),

for the set $\mathcal{S}_3$. Note that $(\log T)^3 \leq V \leq T^{\frac{1}{2}}$ suffices for (46) and that (7), (8) and (9) are trivial if the former condition does not hold.

If $V \geq T^{\frac{1}{2}}$ then

$$|t_m| \gg |\zeta(\tfrac{1}{2} + it_m)| \geq V \geq T^{\frac{1}{2}}.$$

We may therefore suppose, without loss of generality, that $T^{\frac{1}{2}} \leq t_m \leq T$ for the proof of (7), (8) and (9). We cover the interval $[T^{\frac{1}{2}}, T]$ with subintervals $[(\tfrac{1}{2})^n, (\tfrac{1}{2})^{n+1}]$. Then (7), (8) and (9) may be applied to each subinterval, and, on summing for n , we obtain the full result. We note finally that (6) follows from Theorem 10.3 of Montgomery [7], and that (6) implies (7), for $V \leq T^{\frac{1}{2}}$.

To prove Theorem 1 we write

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^{12} dt \leq \sum_{0 \leq m \leq T} \max_{m \leq t \leq m+1} |\zeta(\tfrac{1}{2} + it)|^{12}.$$

Let the maximum over $m \leq t \leq m+1$ occur at t_m . Then $|t_{2m} - t_{2n}| \geq 1$ for $m \neq n$. We write $\mathcal{S}_v$ for the set of t_{2m} for which

$$2^v \leq |\zeta(\tfrac{1}{2} + it_{2m})| \leq 2^{v+1}.$$

Then (7) applies, with $V = 2^v$, and yields

$$\sum_{t_{2m} \in \mathcal{S}_v} |\zeta(\tfrac{1}{2} + it_{2m})|^{12} \ll T^2 (\log T)^{16}.$$

We sum for $0 \leq v \leq 2 \log T$, whence

$$\sum_{0 \leq t_{2m} \leq T+1} |\zeta(\tfrac{1}{2} + it_{2m})|^{12} \ll T^2 (\log T)^{17},$$

since the contribution in case $|\zeta(\tfrac{1}{2} + it_{2m})| \leq 1$ is $O(T)$. Similarly we have

$$\sum_{0 \leq t_{2m+1} \leq T+1} |\zeta(\tfrac{1}{2} + it_{2m+1})|^{12} \ll T^2 (\log T)^{17},$$

and Theorem 1 follows.

The proof of the corollary is similar to that given for Theorem 1, except that we also use (8) and (9), for the ranges of V given in Theorem 2. Noting that

$$2^v \ll T^{\frac{12v}{1067}} (\log T)^{\frac{12v}{1067}}$$

by Kolesnik's estimate (5), we deduce that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^k dt \ll (T^{(3k+4)/20} + T^{1+173(k-6)/1067}) (\log T)^{2k},$$

for $k \geq 12$, and the corollary follows.

REFERENCES

1. F. V. Atkinson, 'The mean value of the Riemann zeta function', *Acta Math.*, 81 (1949), 353–376.
2. R. Balasubramanian, 'An improvement on a theorem of Titchmarsh on the mean square of $|\zeta(\frac{1}{2} + it)|^2$ ', *Proc. London Math. Soc.*, (3) 36 (1978), 540–576.
3. G. H. Hardy and J. E. Littlewood, 'The Riemann Zeta-function and the theory of the distribution of primes', *Acta Math.*, 41 (1918), 119–196.
4. D. R. Heath-Brown, 'The fourth power moment of the Riemann zeta function', to appear.
5. A. E. Ingham, 'Mean-value theorems in the theory of the Riemann zeta-function', *Proc. London Math. Soc.*, (2) 27 (1926), 273–300.
6. G. A. Kolesnik, 'On the estimation of some trigonometric sums', *Acta Arith.*, 25 (1973), 7–30.
7. H. L. Montgomery, *Topics in multiplicative number theory*, Springer, Berlin, 1971.
8. E. C. Titchmarsh, *The theory of the Riemann Zeta-function*, Oxford, 1951.

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